The Pentagon, Machine-Checked: Abel's five-term relation for the dilogarithm in Lean 4

Special functions · the dilogarithm

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Statements

Theorem 1 — Dilog.rogersL_fiveterm 3

At a glance

What the paper contributes

A machine-checked proof of the identity that generates the weight-2 theory of the dilogarithm, on the dilogarithm built in Lean by the companion paper.

- **Abel's five-term relation** (Theorem 1), sorry-free in Lean 4: for x, y in (0,1), L(x) + L(y) = L(xy) + L(x(1−y)/(1−xy)) + L(y(1−x)/(1−xy)), with L the Rogers L-function.
- **The classical route, made formal**: the five-term difference has zero derivative in x (nine logarithms collapse into a basis of five), so it is constant; its value is read off as $x \to 0^+$, where $\log t \cdot \log(1-t) \to 0$ by a squeeze.
- **Three reusable blocks**: the derivative of L on (0,1), $L(t) \to 0$ as $t \to 0^+$, continuity of L on (0,1).
- The mathematics is classical (Spence, Abel, Rogers); the paper claims the formalization, after a documented search of proof-assistant libraries.

🚀 **Start here: Theorem 1**: Abel's five-term relation for the Rogers L-function on $(0,1)^2$.

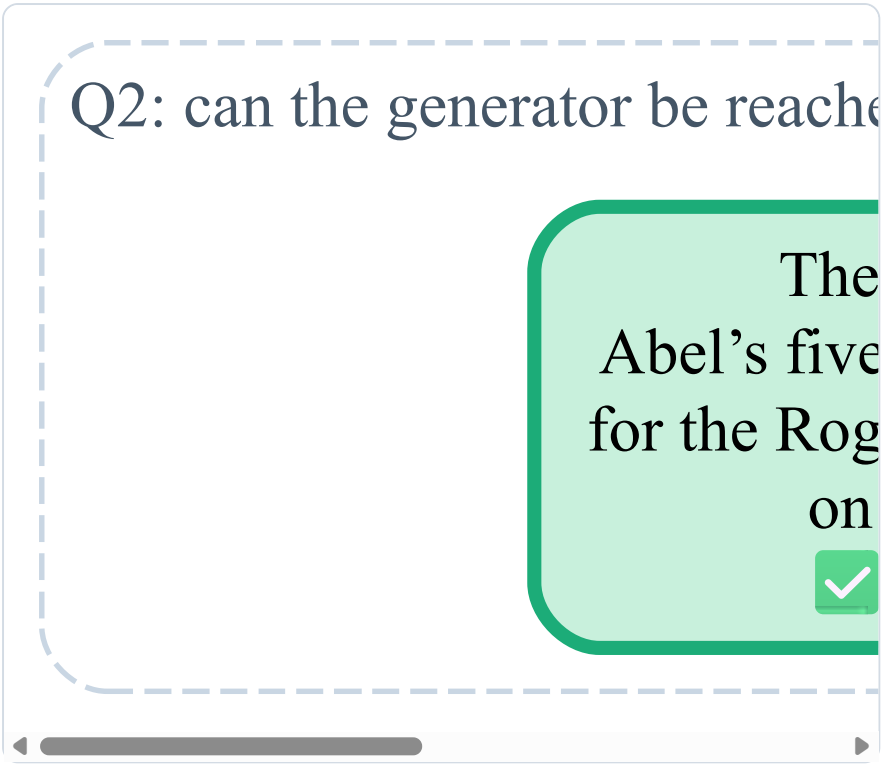
Main results:

- **Theorem 1**: Abel's five-term relation for the Rogers L-function on $(0,1)^2$

1 results | ✅ **1** formalized in Lean | ◐ **0** with a part in Lean | 💻 **0** checked by a program | written proof only | (a result can carry several tags)

🗺️ Map of the results (click a box to go to it)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's "Evidence & scope" block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



📐 **Setting (copied from the paper, §1)**

The cleanest statement uses the *Rogers* L-*function* $\mathrm{L}(x) := \mathrm{Li}_2(x) + \frac{1}{2}\log x \log(1-x)$, which absorbs the stray logarithms of the raw Li_2 form and turns the relation into a pure sum of five L values. *(§2)* Fix $y \in (0,1)$ and set

$$G(t) = \mathrm{L}(t) - \mathrm{L}(ty) - \mathrm{L}\left(\tfrac{t(1-y)}{1-ty}\right) - \mathrm{L}\left(\tfrac{y(1-t)}{1-ty}\right).$$

The theorem is $G(x) = -\mathrm{L}(y)$.

2 Q2: can the generator be reached by the same three-move recipe?

Theorem 1. ✅ Lean (`Dilog.rogersL_fiveterm`)✓ 📕 p. 3

> *For all* $x, y \in (0,1)$,
>
> $$\mathrm{L}(x) + \mathrm{L}(y) = \mathrm{L}(xy) + \mathrm{L}\left(\tfrac{x(1-y)}{1-xy}\right) + \mathrm{L}\left(\tfrac{y(1-x)}{1-xy}\right).$$

✅ Checked when this page was built: FiveTerm.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — rogersL_fiveterm, fiveterm_hasDerivAt_zero, hasDerivAt_rogersL, tendsto_rogersL_nhdsGT_zero, continuousAt_rogersL: propext, Classical.choice, Quot.sound. Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean FiveTerm.lean` (SHA-256 of the compiled file: `ce29fa490a366ddb…`; a source page on this site may show it with comments edited, under its own hash).

> 🆕 **Classical and new:** classical (Spence, Abel, Rogers); the paper claims the formalization: «to the best of my knowledge the first machine-checked five-term relation, on the first machine-checked dilogarithm, in any proof assistant» (§6), from a documented search of June 2026.

↳ Not used by another result of the paper.

Proof

> Formalized in Lean (see the linked declarations).

References

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